Coding(Software)

Senior Category

(**14-19 years** Old)

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STEM FESTIVAL 2026

Official Competition Tasks – Coding (Software)

Category: Senior (14–19 Years)

Theme: Engineering Tomorrow: Agriculture I Education I Health

Track: Agriculture

Focus: Advanced Data Analytics & API Integration

Task Title: The "Agro-Analytics" Farm Dashboard

Team Size: 3 Participants

Governing Documents: STEM Festival Global Rulebook (2026)

Official Reference: www.stemfestival.org

1. OVERVIEW

The Senior Coding Category focuses on high-level data integration. This task challenges teams to build an "Agro-Analytics" Farm Dashboard. This is a comprehensive web or desktop platform that integrates real-time weather data and market prices with farm-specific inventory data, allowing commercial farmers to make data-driven decisions on planting, harvesting, and selling.

2. TASK OPTION 1 – AGRICULTURE

The "Agro-Analytics" Farm Dashboard



Problem Statement

Commercial farmers face huge risks when they lack real-time data on weather and market fluctuations. Relying on "yesterday's news" can lead to crop loss or selling at a loss. We need a Senior-level software solution that centralizes external data (API) and internal farm data into one powerful decision-making cockpit.

Task Guidelines

Teams must develop a functional analytics dashboard that includes:

- **Market Price Integration:** Use of an API or simulated live-feed to display current prices for at least three local crops.
- **Weather Intelligence:** Integration of a weather API to show localized forecasts for the farm's specific coordinate.
- **Inventory Analytics:** A data system that calculates the "Estimated Market Value" of current farm stock based on live prices.

Functional Requirements

- **API Connectivity:** The app must successfully fetch and display data from at least one external source.
- **Data Visualization:** Use of advanced charts (line, bar, or pie) to represent inventory and market trends.
- **Responsive Design:** The dashboard must be accessible and readable across different screen sizes.

3. DELIVERABLES & SUBMISSION INSTRUCTIONS

- **Email Subject Line:** SEN_Coding_Agriculture_TeamName_AgroAnalytics
- **Submission Email:** senior@stemfestival.org

Timeline: All work must be submitted at least 15 days prior to the October 28, 2026 kickoff (**Deadline: October 13, 2026**).

- **Task Overview:** A summary of the APIs used and how the "Estimated Market Value" logic was implemented.
- **Functional Prototype:** A link to the hosted web application or the source code repository (e.g., GitHub).
- **Demonstration Video:** 5–7 minutes showing the live API data fetching and the dashboard's interactive charts.
- **Written Report:** 4–8 pages covering API documentation, database schema, and an analysis of how the dashboard improves farm profit.

4. SUGGESTED TOOLS & TECHNOLOGIES (FREE CHOICE)

Software:

- **Frontend:** React, Vue.js, or HTML5/Bootstrap.
- **Backend/Logic:** Python (Flask/Django), Node.js, or PHP.
- **APIs:** OpenWeatherMap, Agromonitoring API, or simulated REST APIs.
- **Visualization:** Chart.js, D3.js, or Plotly.

Hardware:

- **Standard:** High-performance Laptop or Desktop for development.
- **Server:** Use of a free-tier hosting service (e.g., Vercel, Netlify, or Heroku).

5. SCORING CRITERIA (100 POINTS)

Criterion	Description	Points
API Integration	Success in fetching, parsing, and displaying external live data	25
Analytics Logic	Accuracy of the value calculations and data correlations	20
UI/UX Excellence	Professionalism, layout, and clarity of the dashboard interface	15
Code Structure	Quality of the backend logic, security, and API handling	15
Innovation	Unique features like harvest alerts or historical price trends	15
Documentation	Quality of the technical report and the demo video	10
TOTAL	**Sum of all evaluation points**	**100**

6. CONCLUSION

The "Agro-Analytics" dashboard transforms farming from a gamble into a calculated science. By building this platform, Senior coders are engineering the financial stability and growth of Nigeria's agricultural sector.

7. ETHICAL CODE

- **Data Integrity:** Teams must ensure that all market and weather data displayed is accurate.
- **Security:** If using API keys, teams must practice secure coding (e.g., using environment variables).
- **Transparency:** Clearly state which parts of the data are live and which are simulated.

8. ACKNOWLEDGMENT & AGREEMENT

By participating in this task, teams agree to the rules and standards set forth in the STEM Festival Global Rulebook (2026).

9. DISCLAIMERS

- The system is a decision-support tool; users are responsible for final farm decisions.
- Organizers are not responsible for API downtime during the judging period.

10. GENERAL COMPETITION RULES

- All submissions must be made through the official portal or designated email.
- **The judges' decision is final and binding.**

STEM FESTIVAL 2026

Official Competition Tasks – Coding (Software)

Category: Senior (14–19 Years)

Theme: Engineering Tomorrow: Agriculture | Education | Health

Track: Education

Focus: Mobile Development & Database Integration

Task Title: The "**Exam-Prep**" Mobile Platform

Team Size: 3 Participants

Governing Documents: STEM Festival Global Rulebook (2026)

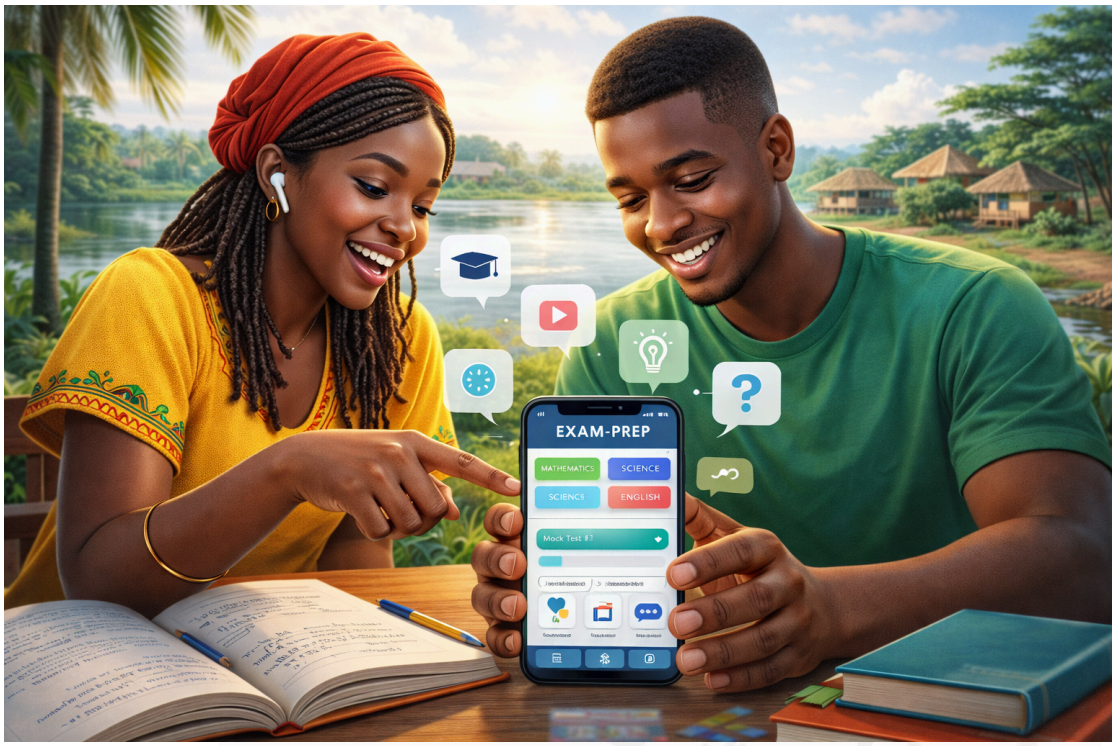
Official Reference: www.stemfestival.org

1. OVERVIEW

The Senior Coding Category for Education addresses the challenge of high-stakes examination preparation. This task requires teams to build a comprehensive mobile application designed to help students prepare for national exams like JAMB and WAEC. The platform must offer more than just text; it must provide timed practice tests, performance analytics, and a centralized repository of study materials.

2. TASK OPTION 2 – EDUCATION

The "Exam-Prep" Mobile Platform



Problem Statement

Many students in Nigeria and in the Niger Delta struggle to access quality preparatory materials for national examinations. Physical past-question books are often outdated or unavailable in rural areas. We need a modern, scalable mobile solution that gives every student—regardless of location—access to high-quality practice tools and real-time feedback on their performance.

Task Guidelines

Teams must develop a functional mobile application that includes:

- **Subject-Based Quizzes:** A database of at least 50 questions across three subjects with multiple-choice logic.
- **Timed Exam Mode:** A feature that simulates real exam conditions by including a countdown timer for quiz sessions.

- **Performance Analytics:** A results screen that shows the user's score and identifies "Weak Areas" based on the questions they missed.

Functional Requirements

- **Database Integration:** The app must retrieve questions and store user scores using a structured database (e.g., Firebase, SQLite, or Airtable).
- **Authentication:** A simple user login/signup system to keep track of individual student progress.
- **Offline Access:** The ability for users to access at least one "Practice Set" without an active internet connection.

3. DELIVERABLES & SUBMISSION INSTRUCTIONS

- **Email Subject Line:** SEN_Coding_Education_TeamName_ExamPrep
- **Submission Email:** senior@stemfestival.org

Timeline: All work must be submitted at least 15 days prior to the October 28, 2026 kickoff (Deadline: October 13, 2026).

- **Task Overview:** A summary of the subjects covered and how the "Weak Area" detection logic works.
- **Functional Prototype:** A link to the GitHub repository or an APK/TestFlight link for the mobile app.
- **Demonstration Video:** 5–7 minutes showing a user registering, taking a timed quiz, and viewing their final analytics.
- **Written Report:** 4–8 pages covering the database schema, UI/UX design wireframes, and a plan for scaling the app to thousands of users.

4. SUGGESTED TOOLS & TECHNOLOGIES (FREE CHOICE)

Software:

- **Frameworks:** Flutter, React Native, or MIT App Inventor (Advanced level).
- **Languages:** Dart, JavaScript, or Python (with Kivy).
- **Backend:** Firebase Authentication and Realtime Database.
- **Assets:** Clean, educational UI kits and royalty-free icons.

Hardware:

- **Standard:** Development PC or Laptop.
- **Testing:** An Android or iOS device to demonstrate mobile performance.

5. SCORING CRITERIA (100 POINTS)

Criterion	Description	Points
Quiz Logic	Accuracy of the scoring, timing, and question-retrieval system	25
Database Performance	Stability of the user authentication and score-storage logic	20
UI/UX Excellence	Professionalism, readability, and ease of navigation for students	15
Mobile Responsiveness	How well the app performs on actual mobile hardware	15
Analytics Value	Usefulness of the feedback provided to the student	15
Professionalism	Quality of the technical report and the demo video	10
TOTAL	**Sum of all evaluation points**	**100**

6. CONCLUSION

The "Exam-Prep" platform levels the playing field for all students. By engineering this digital tutor, Senior coders are ensuring that academic success is determined by effort and intelligence, not by geographical location or wealth.

7. ETHICAL CODE

- **Academic Integrity:** The app must be marketed as a study aid, not a way to cheat.
- **Data Privacy:** Student scores and contact information must be kept secure.
- **Originality:** The codebase must be original; if using templates, the core logic must be customized.

8. ACKNOWLEDGMENT & AGREEMENT

By participating in this task, teams agree to the rules and standards set forth in the STEM Festival Global Rulebook (2026).

9. DISCLAIMERS

- The app is a prototype; teams are responsible for the accuracy of the practice questions.
- Organizers are not responsible for database hosting costs during the competition.

10. GENERAL COMPETITION RULES

- All submissions must be made through the official portal or designated email.
- **The judges' decision is final and binding.**

STEM FESTIVAL 2026

Official Competition Tasks – Coding (Software)

Category: Senior (14–19 Years)

Theme: Engineering Tomorrow: Agriculture I Education I Health

Track: Health

Focus: Full-Stack Development & API Logic

Task Title: The "**Telemedicine Appointment**" System

Team Size: 3 Participants

Governing Documents: STEM Festival Global Rulebook (2026)

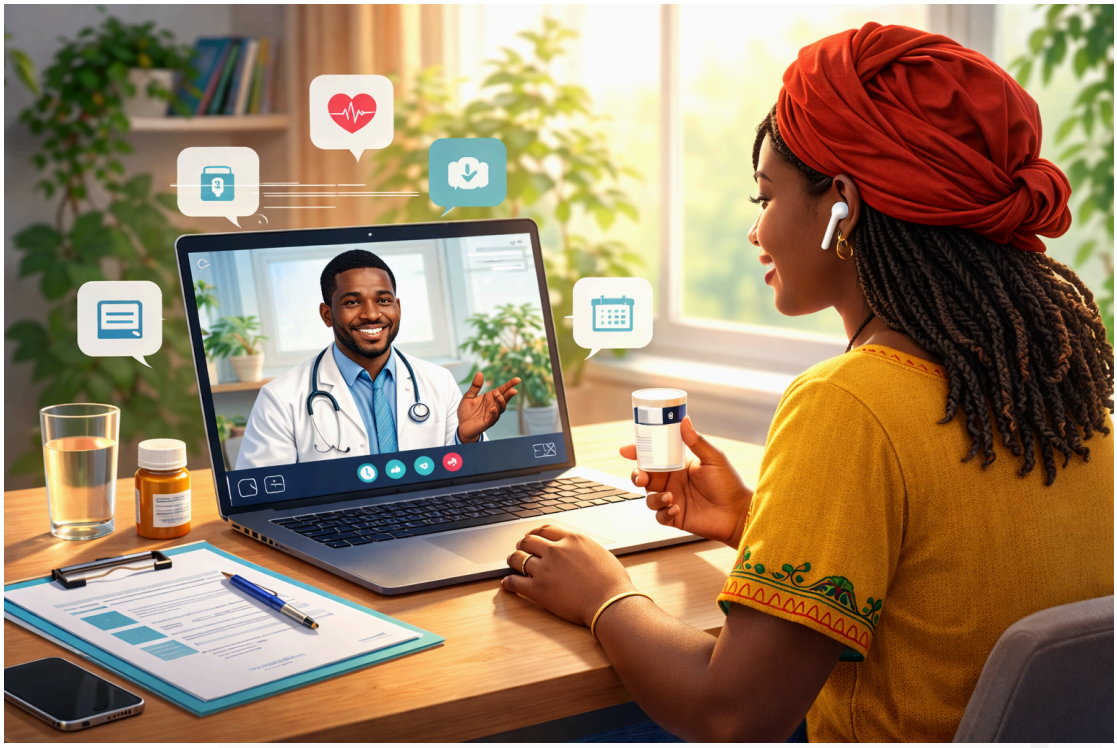
Official Reference: www.stemfestival.org

1. OVERVIEW

The Senior Coding Category for Health tackles the issue of medical accessibility in Africa especially in the Niger Delta. This task challenges teams to build a **Telemedicine Appointment System**. This is a web-based platform that connects patients with doctors, allowing for remote appointment booking, digital health record management, and a secure interface for medical consultations.

2. TASK OPTION 3 – HEALTH

The "Telemedicine Appointment" System



Problem Statement

In many parts of Nigeria, visiting a hospital involves long travel times and hours spent in waiting rooms. This prevents many people from seeking early medical advice. We need a software solution that streamlines the "Doctor-Patient" relationship, allowing for efficient scheduling and remote communication to reduce the burden on physical hospital infrastructure.

Task Guidelines

Teams must develop a functional web-based system that includes:

- **Doctor Availability Calendar:** A feature where doctors can set available times and patients can book specific slots.
- **Digital Health Profile:** A secure section for patients to upload and view their basic medical history and current symptoms.
- **Booking Notification System:** Automatic feedback (visual or email-simulated) confirming the appointment time to both the doctor and patient.

Functional Requirements

- **User Roles:** The system must distinguish between a "Doctor" login (who sees the schedule) and a "Patient" login (who books).
- **Data Security:** Implementation of basic encryption or password protection for patient profiles.
- **Scheduling Logic:** Code to prevent "Double Booking" of the same time slot.

3. DELIVERABLES & SUBMISSION INSTRUCTIONS

- **Email Subject Line:** SEN_Coding_Health_TeamName_Telemedicine
- **Submission Email:** senior@stemfestival.org

Timeline: All work must be submitted at least 15 days prior to the October 28, 2026 kickoff (Deadline: October 13, 2026).

- **Task Overview:** A summary of the scheduling logic and the security measures implemented for patient data.
- **Functional Prototype:** A link to the hosted website (e.g., Vercel or Heroku) and the source code repository.
- **Demonstration Video:** 5–7 minutes showing a patient booking an appointment and a doctor viewing that appointment on their dashboard.
- **Written Report:** 4–8 pages covering the system architecture, the database model, and a discussion on medical data privacy.

4. SUGGESTED TOOLS & TECHNOLOGIES (FREE CHOICE)

Software:

- **Frontend:** React, Angular, or Vue.js.
- **Backend:** Node.js, Django, Flask, or Laravel.
- **Database:** PostgreSQL, MySQL, or MongoDB.
- **Styling:** CSS Frameworks like Tailwind or Bootstrap.

Hardware:

- **Standard:** High-performance Laptop or Desktop for development and hosting.

5. SCORING CRITERIA (100 POINTS)

Criterion	Description	Points
System Logic	Accuracy of the scheduling, role management, and booking logic	25
Data Security	Effectiveness of the measures taken to protect patient privacy	20
Full-Stack Execution	How well the frontend and backend communicate and sync data	15
UI/UX Excellence	Professionalism and ease of use for both doctors and patients	15
Complexity	Use of advanced features like email simulations or file uploads	15
Professionalism	Quality of the technical report and the demo video	10
TOTAL	**Sum of all evaluation points**	**100**

6. CONCLUSION

The "Telemedicine Appointment" system brings the doctor's office to the patient's home. By building this platform, Senior coders are engineering a more efficient, accessible, and modern healthcare system for everyone.

7. ETHICAL CODE

- **Confidentiality:** Patient data must be treated as strictly private.
- **Safety:** The system must clearly state it is for non-emergency appointments only.
- **Honesty:** Teams must be transparent about the limitations of their prototype.

8. ACKNOWLEDGMENT & AGREEMENT

By participating in this task, teams agree to the rules and standards set forth in the STEM Festival Global Rulebook (2026).

9. DISCLAIMERS

- This system is a prototype; it must not be used for actual medical diagnosis or treatment.
- Teams are responsible for securing their web servers against unauthorized access.

10. GENERAL COMPETITION RULES

- All submissions must be made through the official portal or designated email.
- **The judges' decision is final and binding.**